



Quantum Cinema

Making the Invisible Visible Through Generative World Models

Quantum Cinema turns hidden quantum hardware into a cinematic browser experience. It combines generative world models, AWS Braket device metrics, and a guided four-step interface to support science communication and academic reproducibility.

[View on GitHub](#)

[Supplementary Material](#)

The Imagination Gap

The Problem

Quantum computers promise to reshape medicine, climate science, and cryptography — yet they remain locked behind layers of abstraction: hidden in refrigerators, understood only by physicists, visually indistinguishable from sculpture.

The 2025 Nobel Prize in Physics — awarded to **John Clarke, Michel Devoret, and John Martinis** for demonstrating macroscopic quantum tunnelling in superconducting circuits — made this gap urgent.

What Makes It "Quantum"?

Classical computers use **bits** (0 or 1). Quantum computers use **qubits**, exploiting two phenomena:

- **Superposition** — a qubit can be 0 and 1 simultaneously until measured, like a spinning coin
- **Entanglement** — two qubits linked so measuring one instantly determines the other ("spooky action at a distance")
- **Decoherence** — quantum states are fragile; heat or vibration collapses them back to classical 0 or 1

The Complete Journey

From a historic Nobel Prize to hands-on understanding of why building a quantum computer is so hard.

1

Act 01 — Nobel Prize

"Why does this matter — and why now?" A 125-year timeline from Planck (1900) to Quantum Supremacy (2019) to the 2025 Nobel.

2

Act 02 — World Models

"Watch the invisible become visible." AI-dreamed documentary clips of three quantum architectures. Pick the machine you want to explore.

3

Act 03 — Explore

"Step inside the machine you chose." An explorable World Labs 3D scene plus a guided breakdown of entanglement in that architecture.

4

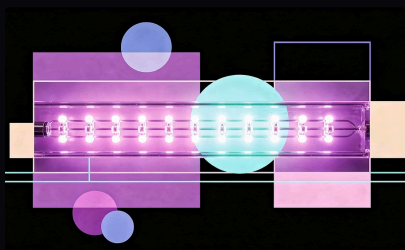
Act 04 — Compare

"Why is there no single best quantum computer?" Radar chart and metric tables across all three devices reveal the trade-offs.



Three Quantum Architectures

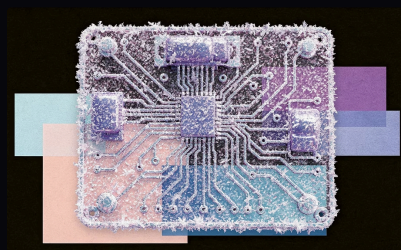
Each quantum architecture on AWS Braket is reimagined as a named visual world — with real performance metrics and a three-beat documentary narrative revealing both its superpower and its Achilles' heel.



🌌 IonQ Aria — "Light Suspension"

Ytterbium ions levitated in a vacuum, pinned by light, holding quantum state for **seconds** — an eternity in the quantum world.

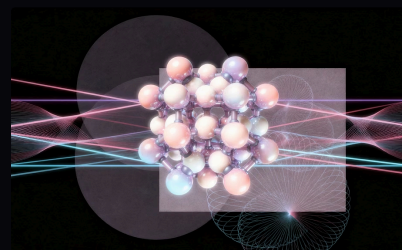
- Coherence: ~1–10 s
- Fidelity: 99.5%+
- **Catch:** Gates are agonizingly slow



⚡ Rigetti Ankaa-3 — "Frozen Forge"

Golden superconducting circuits encased in frost, colder than the void between stars, firing billions of operations before the cold loses its grip.

- Coherence: ~20–100 μ s
- Fidelity: 99.0%+
- **Catch:** Coherence lasts only microseconds



🌊 QuEra Aquila — "Wave Garden"

Neutral atoms arranged by optical tweezers into programmable geometries — nature's own quantum simulator, solving physics by being physics.

- Coherence: ~1–10 μ s
- Fidelity: ~97–99%
- **Catch:** Analog only; can't run general digital algorithms

The Three Invisible Forces

Quantum Cinema makes three physical phenomena — the forces that decide whether a quantum computer is a revolutionary tool or an expensive paperweight — **observable as visual narrative**.

Decoherence

Qubits losing their quantum state to environmental noise.

Rendered as: Frost creeping over circuits; light fading from suspended ions.

Laser Cooling

Bombarding atoms with light to slow them to microkelvins.

Rendered as: The humming, power-hungry glow of laser arrays vs. the stillness they create.

Energy Loss

Heat dissipation destroying quantum information.

Rendered as: Golden circuitry leaking light into the surrounding dark.

What Decides Everything

The core lesson of the Comparison act: **you cannot optimize all six at once**. Every architecture trades some away to win others — and that trade-off is exactly why "which quantum computer is best?" is the wrong question.

Coherence Time

How long a qubit stays "quantum" before noise destroys its superposition. Short coherence = the simulation crashes before finding the answer.

Gate Fidelity

Accuracy of each operation. 99.9% means 1 error per 1,000 operations; errors accumulate into nonsense.

Connectivity

How many other qubits each qubit can directly "talk to." Limited connectivity forces slow, wasteful workarounds.

Error Rate

Above ~1%, error correction can demand 1,000+ physical qubits per usable logical qubit.

Energy Cost

Power to maintain the quantum state — lasers for ions, cryogenics for superconductors.

Qubit Count

Raw scale, which only matters when paired with the five qualities above.



How It's Built

A single Next.js 16 application running inside a container on AWS. CloudFront sits in front for speed and security, the container is hidden in a private network, and the immersive 3D worlds are streamed from World Labs. No database, no quantum hardware in the request path.

01

CloudFront (CDN)

Enforces HTTPS, HSTS, security headers, and caches heavy video/static assets globally.

02

Application Load Balancer

Secret-header gate: forwards traffic only if the CloudFront secret header is present. All else → 403.

03

ECS Fargate (Private Subnets)

Next.js 16 standalone container, Node 20 Alpine, non-root. Auto-scales 1→4 tasks at 70% CPU.

04

World Labs (marble.worldlabs.ai)

Generative 3D worlds — one per quantum architecture — streamed in a new tab from the Explore step.

✨ Key Features & Technology

Key Features



Four-Act Cinematic Flow

Nobel Prize → World Models → Explore → Compare, with animated step transitions.



Interactive Radar Comparison

Normalized 0–100 scoring across six axes with a live radar chart for all three devices.



Dark / Light Theme

Cinematic dark default with one-tap toggle; no-flash, preference persisted.

Tech Stack

- **Framework:** Next.js 16 (App Router) • React 19
- **Styling:** Tailwind CSS 4 • shadcn/ui
- **Animation:** Framer Motion 12
- **3D Worlds:** World Labs generative scenes
- **Infrastructure:** AWS CDK (TypeScript) • ECS Fargate • ALB • CloudFront • VPC • S3
- **Runtime:** Node 20 Alpine • multi-stage Docker • non-root container



Research Questions & Reproducibility

Research Questions

1

RQ1

Can generative world models render real quantum-device characteristics (coherence, fidelity, connectivity) as perceptually legible cinematic experiences?

2

RQ2

Does a *Choose* → *Watch* → *Explore* → *Compare* flow build quantum intuition in audiences with no technical background?

3

RQ3

Can a browser-only experience — zero install, zero hardware — meaningfully close the imagination gap between quantum impact and public understanding?

Academic Reproducibility

This repository is structured for academic review with a dedicated guide, CI validation, and citation metadata.

- `SUPPLEMENTARY.md` — reproducibility and provenance guide
- `CITATION.cff` — citation metadata for academic use
- `.github/workflows/ci.yml` — automated frontend and infrastructure validation
- `design/design.md` — full research and design rationale

i The package is intended as a supplementary artifact for publication, with reproducible build/deployment workflow and explicit academic citation metadata.

FUTURE RESEARCH



What Comes Next

Quantum Cinema is a **deployed foundation** for accessible quantum science communication. Three research directions it opens:



World Models

Live Braket data → conditioned 4D scene generation. Physically accurate decoherence simulations driven by real hardware metrics.



Science Communication

User studies: do the worlds build quantum intuition? Longitudinal quantum-literacy measurement across diverse audiences.



Live Quantum

Real-time AWS Braket job results driving the visuals. Physical QPU feeds from IonQ, Rigetti, and QuEra on demand.

Making the invisible visible — one frame at a time. Built with generative world models • AWS CDK • real quantum-device data • Next.js 16.
Licensed under the MIT License.

[Explore the Repository](#)

[Quick Start](#)